

Python Programming Assignment – 1

1. Write a program to display:

- Value
- Type
- Memory address

for a variable using appropriate built-in functions.

Answer :

```
"""
1. Write a program to display:
• Value
• Type
• Memory address
for a variable using appropriate built-in functions.
"""

Value1 = 9
Value2 = "AMAR"
Value3 = 3.14

print("Value1 is : ", Value1)
print("Value1 Type is : ", type(Value1))
print("Value1 Memory Address is : ", id(Value1))

print("\nValue2 is : ", Value2)
print("Value2 Type is : ", type(Value2))
print("Value2 Memory Address is : ", id(Value2))

print("\nValue3 is : ", Value3)
print("Value3 Type is : ", type(Value3))
print("Value3 Memory Address is : ", id(Value3))
```

OUTPUT :

```
Value1 is : 9
Value1 Type is : <class 'int'>
Value1 Memory Address is : 140737329677688

Value2 is : AMAR
Value2 Type is : <class 'str'>
Value2 Memory Address is : 1821782134096

Value3 is : 3.14
Value3 Type is : <class 'float'>
Value3 Memory Address is : 1821781908592
```

2. What is the difference between:

a = 10

b = 10

and

a = [10]

b = [10]

Explain using id().

Answer :

```
"""
2. What is the difference between:
a = 10
b = 10
and
a = [10]
b = [10]
Explain using id().
"""

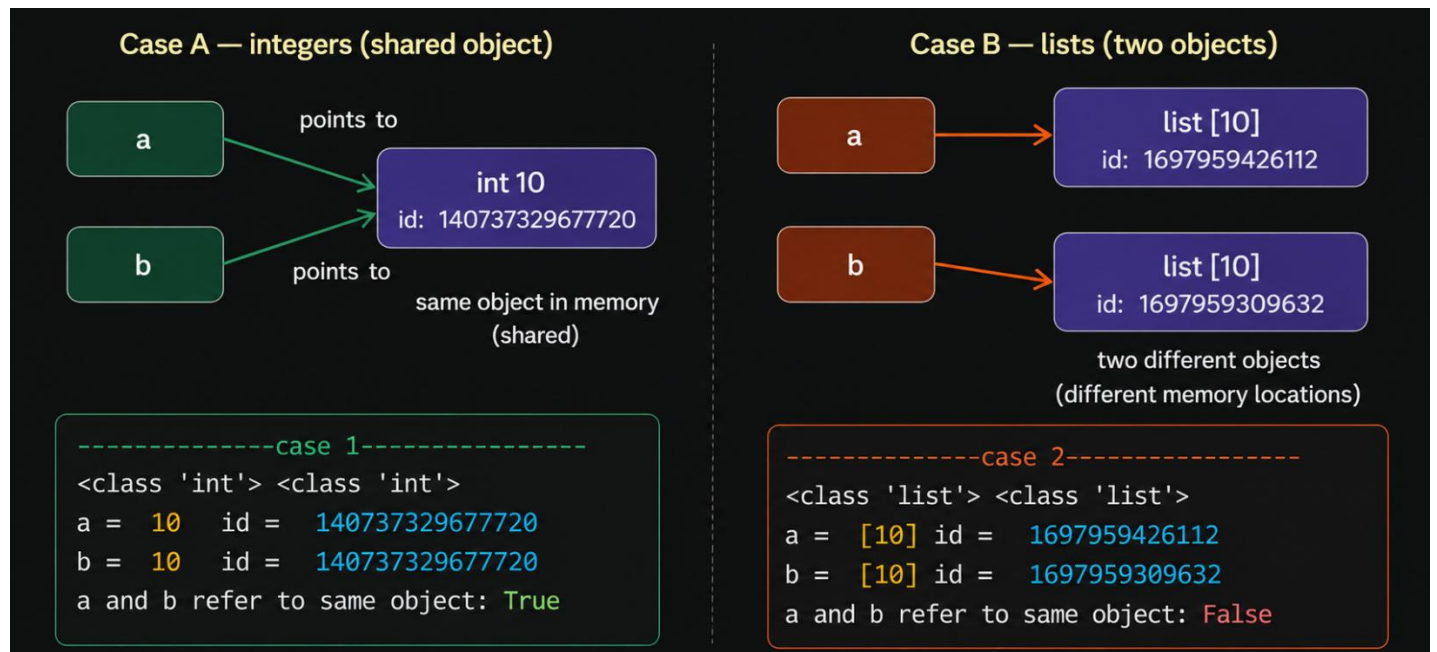
print("-----case 1-----")
a = 10
b = 10
print(type(a), type(b))
print("a = ", a,"id = ",id(a))
print("b = ", b,"id = ",id(b))
print("a and b refer to same object:", id(a) == id(b))      #if id match
return bool True value

print("\n-----case 2-----")
a = [10]
b = [10]
print(type(a), type(b))
print("a = ", a,"id = ",id(a))
print("b = ", b,"id = ",id(b))
print("a and b refer to different object:", id(a) == id(b))

OUTPUT :
-----case 1-----
<class 'int'> <class 'int'>
a = 10 id = 140737329677720
b = 10 id = 140737329677720
a and b refer to same object: True

-----case 2-----
<class 'list'> <class 'list'>
```

```
a = [10] id = 1697959426112
b = [10] id = 1697959309632
a and b refer to same object: False
```



3. What does the id() function return?

Is the value returned by id() same for two variables holding the same value?

Solution :

```
"""
3. What does the id() function return?
Is the value returned by id() same for two variables holding the same
value?
"""

#Declaration
age = 11
marks = 21
no = 11

print("\n----Step 1 : See the variable and object id(memory address) and
type----")
print("Age is :", age)
print("Type is : ", type(age))
print("Address is : ", id(age))

print("Marks is :", marks)
print("Type is : ", type(marks))
```

```

print("Address is : ", id(marks))

print("No is :", no)
print("Type is : ", type(no))
print("Address is : ", id(no))

print("\n-----Step 2 : update no = 11 to 9.8 and see the changes----")
no = 9.8
print("Age is :", age)
print("Type is : ", type(age))
print("Address is : ", id(age))

print("Marks is :", marks)
print("Type is : ", type(marks))
print("Address is : ", id(marks))

print("No is :", no)
print("Type is : ", type(no))
print("Address is : ", id(no))

print('\n Note : \n We see After updation if object available in another
then point to that location otherwise allocate new memory and here
dynamically variable type can change')

print("\n-----Step 3 : update age = 11 to 12 and see the changes----")
age = 12
print("Age is :", age)
print("Type is : ", type(age))
print("Address is : ", id(age))

print("Marks is :", marks)
print("Type is : ", type(marks))
print("Address is : ", id(marks))

print("No is :", no)
print("Type is : ", type(no))
print("Address is : ", id(no))

print('\n Note : \n Here we see if we update objects value and no one point
that memory address mean references count is zero then memory was released
and updated value store in new address place')

```

4. What is the purpose of `getsizeof()`?

Why is memory size different for different data types?

Answer :

```
import sys

x = 10
y = 3.14
name = "Python"

print(sys.getsizeof(x))
print(sys.getsizeof(y))
print(sys.getsizeof(name))
print("\n\n")

print(sys.getsizeof(10))
print(sys.getsizeof(3.14))
print(sys.getsizeof(True))
print(sys.getsizeof("Python"))
print(sys.getsizeof([10,20,30]))
```

The `getsizeof()` function is used to determine the **memory size (in bytes)** occupied by an object in Python. It is available in the `sys` module.

Different data types require different amounts of memory because they store different kinds and amounts of information.

5. Predict the output:

```
a = 10
b = 10
print(id(a) == id(b))
```

Explain **why** this happens.

Answer :

```
"""
5. Predict the output:
a = 10
b = 10
print(id(a) == id(b))
Explain why this happens.
"""

a = 10
b = 10

print(id(a))
print(id(b))
```

```
print(id(a) == id(b))
```

OUTPUT :

```
140736772687256
```

```
140736772687256
```

```
True
```

Output: True

Both a and b refer to the same integer object 10. Therefore, id(a) and id(b) return the same identity, making the expression id(a) == id(b) evaluate to True. Python reuses immutable objects like small integers to optimize memory usage.

6. Write a program that accepts two numbers from user and prints their:

- Addition
- Subtraction
- Multiplication
- Division

Answer :

```
"""
6. Write a program that accepts two numbers from user and prints their:
• Addition
• Subtraction
• Multiplication
• Division
"""

# Accept two numbers from the user

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

# Perform arithmetic operations

addition = num1 + num2
subtraction = num1 - num2
multiplication = num1 * num2
division = num1 / num2

# Display results

print("\nAddition =", addition)
print("Subtraction =", subtraction)
print("Multiplication =", multiplication)
print("Division =", division)
```

```
OUTPUT :
Enter first number: 12
Enter second number: 10

Addition = 22.0
Subtraction = 2.0
Multiplication = 120.0
Division = 1.2
```

7. Why does the input() function always return a string? How can you convert it to another data type?

Answer :

The input() function is designed to accept data entered from the keyboard. Since keyboard input is received as text, Python returns the entered value as a **string (str)** by default. Because Python performs **string concatenation**, not addition. Here, the string returned by input() is converted into an integer using int().

```
age = int(input("Enter your age: "))

print(age)
print(type(age))

num = int(input("Enter a number: "))
print(num + num)
print(type(num))
```

```
OUTPUT :
Enter your age: 25
25
<class 'int'>

Enter a number: 234
468
<class 'int'>
```

8. Predict the output:

```
x = input("Enter number: ")
print(type(x))
```

Explain the reason.

Answer :

```
"""
8. Predict the output:
```

```
"""  
  
x = input("Enter number: ")  
print(type(x))
```

OUTPUT :
Enter number: 23
<class 'str'>

9. Write a program to take user's name and age and display: Hello <name>, you will turn <age+1> next year.

Answer :

```
"""  
9. Write a program to take user's name and age and display:  
Hello <name>, you will turn <age+1> next year.  
"""  
  
name = "AMAR"  
age = 28  
  
print("Hello", name, ",", "you will turn", (age+1), "next year.")
```

10.What will be the output: print("10" + "20") print(10 + 20) Explain the difference.

Answer :

```
"""  
10.What will be the output:  
print("10" + "20")  
print(10 + 20)  
Explain the difference  
"""  
  
print("10" + "20")          #take as a str  #1020  
print(10 + 20)              #it is integer  #30
```